

Zinan Xiao

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EDUCATION

King Abdullah University of Science and Technology

Doctor of Philosophy in Electrical & Computer Engineering

Aug. 2024 – May 2028

Jeddah, Saudi Arabia

- **Coursework:** Integrated Sensors, Semiconductor Optoelectronic Devices

Imperial College London

Master of Science in Optics and Photonics

Oct. 2020 – Oct. 2021

London, United Kingdom

- **Coursework:** Optical Design, Laser Optics, Optical Communications, Nonlinear Optics, Lasers Fundamentals

Chongqing University

Bachelor of Engineering in Optoelectronic Information Engineering

Sep. 2016 – Jun. 2020

Chongqing, China

- **Coursework:** Optoelectronic Devices, Principle of Laser, Semiconductor Physics, Optoelectronics Technology

EXPERIENCE

Huawei Technologies Company Limited

Optical Hardware Engineer

Mar. 2022 – May 2023

Wuhan, China

- Improved the productivity of Bragg grating by **20%** and reduced the cost of optical splitter by **40%**, by building the system with team to achieve writing Bragg gratings on ribbon fibers
- Increased the amount of PLC chips by **10%** for wafer with fixed size, by designing dense PLC chips and optimizing the precision of cutting wafer
- Ensured that the 1:5 optical splitter with dense PLC chips can be appropriately used in fiber-to-the-room (FTTR) solution (around **10000** pieces per year), by accomplishing reliability tests such as HAST

PROJECTS

NCPD Date Fruit Types and Maturity Classification Project

- Collected date fruit images in real field with camera and classified and labelled these images in different groups
- Built convolutional neural network (CNN) and achieved **90%** accuracy for classification of 5 different types of date fruit

Aramco Ghost Imaging Project

- Built ghost imaging system in both free space and fiber system, reconstructing different targets with high quality
- Performed convolutional neural network with reconstructed images and compared the denoised images with the benchmark, achieving peak signal-to-noise ratio (PSNR) values up to **12.79 dB** for complex experimental targets

PUBLICATIONS

- [1] Z. Xiao et al., "High-resolution ghost imaging denoising using simulation-based deep learning," *Opt. Express*, vol. 34, no. 5, pp. 7362–7377, 2026. doi: 10.1364/OE.585942. [Online]. Available: <https://opg.optica.org/oe/abstract.cfm?URI=oe-34-5-7362>.
- [2] Z. Xiao et al., "Ghost imaging through multimode fiber with deep learning denoising," in *Optical Fiber Communication Conference*, Optica Publishing Group, 2026, W2A–21.

SKILLS

Languages : Python, MATLAB, C++,

Tools : Zemax, GitHub, Anaconda, VScode